

Drug-Target Interactions Prediction Using Stacking Ensemble Learning Approach

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Abstract—The process of drug discovery, particularly in the domain of drug-target interactions (DTI), is often time-consuming and costly, requiring extensive experimentation and validation before global approval can be obtained. To streamline this process and reduce associated costs, computational methods such as machine learning and deep learning are increasingly employed. However, these approaches often face challenges, including the need for large datasets, significant computational resources, and a tendency towards overfitting. Addressing these limitations, this study explores the use of Stacking Ensemble Learning (SEL) as a promising solution for DTI prediction. The proposed SEL model integrates multiple base learners, including Adaptive Boosting, Gradient Boosting, K-Nearest Neighbor, Random Forest, and Support Vector Machine, and enhances their performance through careful parameter tuning. To address data imbalances, the Synthetic Minority Oversampling Technique (SMOTE) is employed, ensuring more reliable predictions. The model was rigorously evaluated and demonstrated remarkable efficacy, achieving an accuracy of 99.047%. This research underscores the potential of the SEL approach in advancing computational drug discovery by enhancing prediction accuracy and robustness, offering a viable pathway for more efficient and cost-effective drug development processes.

Keywords—Accuracy, Drug-Target Interaction, Prediction, SMOTE, Stacking Ensemble Learning

I. INTRODUCTION

Drug discovery involves the exploration of novel drug candidates, encompassing both chemical and biological avenues. A crucial aspect within this field is the investigation of drug-protein interactions (DTI). Recent advancements in medical technology have enabled deeper and more extensive research into DTI [1]. However, due to the vast array of drug compounds and target proteins, many of these interactions remain undiscovered. Enhancing our understanding of DTI not only aids in comprehending drug pharmacology better but also proves significant in predicting side effects and facilitating drug repositioning—utilizing existing drugs to treat previously unaddressed ailments.

In reality, only a small number of new drugs receive approval from United States Food and Drug Administration (FDA) each year. Additionally, the average expenditure associated with discovering new drugs stands at around \$1.8 billion, and the entire process spans over a decade [2]. Predicting drug-target interactions (DTI) holds significance in the scope of drug discovery, offering potential reductions in both costs and time associated with drug development. Hence, there is a necessity for computational techniques to forecast DTI, aiming to streamline processes, minimize expenses, and conserve experimental resources [3].

A variety of computational methodologies, ranging from machine learning, deep learning to ensemble learning, are

employed in making DTI predictions. Both machine learning (ML) and deep learning (DL) exhibit weaknesses; ML struggles with large datasets, while DL is susceptible to overfitting. Ensemble learning (EL) involves the combination of different learning models to enhance the outcomes derived from each individual model. Among the varieties of EL, stacking ensemble learning (SEL) stands out. SEL addresses the limitations of both ML and DL by efficiently handling big data, improving generalization performance thereby reducing overfitting and bias, and offering enhanced interpretability. Many studies have employed SEL, for the example was research conducted by [4] focusing on predicting loan risk in peer-to-peer (P2P) lending platforms. The findings demonstrate remarkable accuracy, exceeding 90% across two distinct datasets. Another one is the study conducted by [5] focusing on forecasting fatalities resulting from earthquakes. The outcomes illustrate that SEL can indeed enhance model performance.

To date, stacking ensemble learning (SEL) has primarily been utilized in the domain of protein-protein interactions (PPI) within drug discovery, with no prior research applying it to drug-target interaction (DTI) predictions. This paper focuses on exploring the utilization of SEL for predicting DTI pairs. In doing so, this study adds to the field by showcasing the effectiveness of SEL in enhancing DTI prediction. Our objective is to provide insights that may contribute to the development of safe drugs for human use.

II. RELATED WORK

A. Ensemble Learning

Sharma & Rani [6] introduced an ensemble framework that incorporates active learning (AL) and dimensionality reduction. AL is employed to enhance the effectiveness of ensemble methods like bootstrap aggregation (bagging) in addressing under-sampling issues, while dimensionality reduction is utilized to manage high-dimensional datasets. The dataset comprises raw drug-target interactions alongside features associated with each drug and protein. Decision Trees (DT) are chosen as the base learner by the researchers. The introduced method outperformed three other methods, achieving respective AUC values of 0.927, sensitivity of 0.886, specificity of 0.864, and G-mean of 0.874.

Thafar et al. [7] proposed a boosting-based network embedding and ensemble method. The dataset consists of protein sequences and drug chemical structures. From this dataset, the DTi2Vec model is applied, starting from building a heterogeneous network based on similarity, then generating features from each drug and protein node using the node embedding technique. The model performance was evaluated for its AUPR value using two ensemble learning classifiers, namely AdaBoost and XGBoost. There are datasets that have high AUPR values with AdaBoost tuning, there are also

datasets that have high AUPR values with XGBoost tuning. Following the comparison with five other methods, DTi2Vec consistently yielded the best performance across all datasets examined.

PreDTI approach presented by Mahmud et al. [8] is for conducting DTI predictions. Their methodology involves utilizing datasets comprising the chemical structures of drugs and protein sequences. They employed various feature extraction techniques, followed by data balancing using the FastUS (Fast Under Sampling) algorithm and feature selection through the MoIFS (Modified Incremental Feature Selection) algorithm. They used LightGBM (Light Gradient Boosting Machine) as the classification model. Comparisons were made between the proposed method and three alternative methods concerning feature selection and classifier, with the PreDTI method demonstrating the highest accuracy.

B. Maintaining the Integrity of the Specifications

Charoenkwan et al. [9] introduced the SPIDER (Stacked Predictor of Druggable pRoteins) method to improve the prediction accuracy of druggable proteins. The dataset consists of a list of druggable and non-druggable proteins. The dataset is trained with ten feature encodings and six algorithms. SEL is used so that test data can be used, and the model can be interpreted. The findings indicate that SPIDER outperforms the base model in both cross-validation and independent tests, showcasing the best results.

Gao et al. [10] proposed the SELPPI (Stacked Ensemble Learning for Protein-Protein Interaction) method to predict new modulators targeting protein-protein interactions (PPI). The dataset used is pdCSM (Protein Data Bank-Consensus Scoring Measures) which consists of 4965 PPI modulators targeting 51 different PPIs. Six algorithms are used as base learners. The six algorithms are then used alternately as a meta learner. SEL is used to maximize the use of various machine learning models, especially the use of features and their combinations. The results are categorized into classification and regression, showing the best performance when compared with the six basic models.

Several DTI studies utilize diverse datasets. Shang et al. [3] and Yao et al. [11] incorporate multiple datasets sourced from various websites, encompassing drug structures and their interactions, drug-disease associations, drug side effects, as well as protein sequences and interactions. He et al. [12], Yuan et al. [13] expand on these datasets by adding drug-drug interactions, protein-protein interactions, and target-disease associations, while excluding drug structures and protein sequences. On the other hand, Dou et al. [14], Lian et al. [15], Hu et al. [16], Song et al. [17], Chen et al. [18] and Mahdaddi et al. [19] solely rely on drug structures and protein sequences as their primary datasets.

Building upon the prior study discussed, this research fills a crucial void by implementing stacking ensemble learning for DTI prediction. The dataset comprises lists of drugs and proteins, with positive and negative pairs denoted as approved or rejected, respectively. To address class imbalance, we proposed the synthetic minority oversampling technique (SMOTE) [20]. Classification is executed by integrating five classifiers—AdaBoost [21], GBoost [22], KNN, RF, and SVM [23]—as base learners, with MLP [24] serving as the meta learner.

III. METHODOLOGY

In this study, DTI analysis uses several steps, namely data acquisition, preprocessing data, implementing stacking ensemble learning, and evaluation. The flowchart can be shown in Fig. 1.

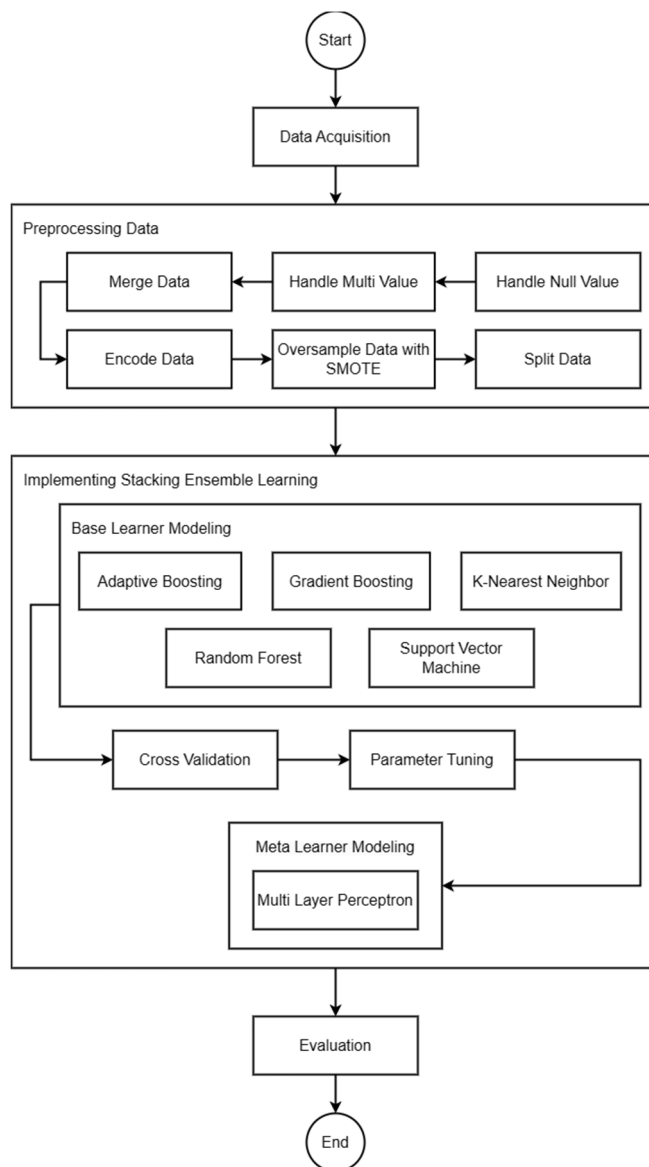


Fig. 1. Methodology Flowchart

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A. Data Acquisition

The initial step of the research methodology involved the acquisition of data through accessing the DrugBank database via the link go.drugbank.com and then downloading the dataset needed. This dataset covered two distinct categories of data: drug and protein data. Within each category, there are two subcategories: approved and rejected entries. Table I and II below is an example showcasing the information provided for each approved drug and protein entry:

TABLE I. APPROVED PROTEIN DATA

Column	Non-Null Count
ID	5346 non-null
Name	5346 non-null
Gene Name	4954 non-null
...	...
HGNC ID	3343 non-null
Species	5244 non-null
Drug IDs	5346 non-null

TABLE II. APPROVED DRUG DATA

Column	Non-Null Count
DrugBank ID	15235 non-null
Name	15235 non-null
CAS Number	9553 non-null
...	...
ChemSpider ID	11480 non-null
BindingDB ID	4979 non-null
TTD ID	1423 non-null

B. Preprocessing Data

The next step is preprocessing data. This entails removing the identifier columns from each protein and drug dataset, involving both approved and rejected entries, leaving behind only one identifier column. The Drug IDs column in the protein dataset contains multiple values, so the data is reshaped to form rows of data with singular values. Then, all protein and drug datasets are merged into a single dataset using the Drug ID column as the basis. Following this, the drug ID is eliminated, and encoding is applied to all columns to facilitate the implementation of stacking ensemble learning.

Significant difference amount of experiment result data values were observed. To address this imbalance, we applied the oversampling technique called SMOTE before implementing the stacking ensemble learning model. SMOTE generates synthetic samples by analyzing the data within the existing minority class. This method generates a synthetic sample by creating linear combinations of two samples from the minority class (x_i and x_j) as shown in equation (1):

$$x_{new} = (x_j - x_i) * \alpha \quad (1)$$

where x_{new} is a newly generated artificial instance. To create the new instance of the minority class (x_{new}), a sample x_i is randomly chosen. Subsequently, x_j is randomly selected from the five nearest neighbors of x_i within the minority class, determined based on the Euclidean distance. α represents a random floating-point value within the range (0, 1) [25]. Then, the dataset was splitted into training and testing data.

C. Implementating Stacking Ensemble Learning

After the preprocessing data step was carried out, next is stacking ensemble learning implementation. Stacking involves combining multiple models of different types through a meta-learner concept [26]. Initially, training data is used to train base learners in the first layer, generating a new dataset, which is then employed to train the second layer meta-learner. The output from the first learner serves as an input feature, while the original label remains the new training data

label. Various classification algorithms are employed by the first-level base learners [27]. The integration of a meta-learner improves the classification accuracy of the stacking model [28] [29].

Let $S = \{(y_n, x_n), n = 1, \dots, N\}$ denote the dataset, where x_n represents the feature vector of the n -th sample, and y_n represents the predicted value for the n -th sample. P corresponds to the number of features, indicating that each feature vector is represented as $(x_1, x_2, \dots, x_P)$. The data is randomly divided into K subsets: $S_1, S_2, \dots, S_K$, each of which is approximately of the same size. Let S_k represent the k -th subset, and S_{-k} denotes the dataset excluding S_k . S_k is defined as the test set, while S_{-k} is defined as the training set for the k -th fold in K-fold cross-validation. In the first layer, the training set S_{-k} is utilized to train the k -th algorithm, resulting in the base model L_k , where $k = 1, 2, \dots, K$.

During K-fold cross-validation, the prediction of sample x_n in the k -th fold test set S_k is denoted as z_{kn} . The value of K is predetermined as 5. Upon completion of the cross-validation procedure, the outputs of the K base learners combine to form new samples:

$$S = \{(y_n, z_{1n}, \dots, z_{Kn}), n = 1, \dots, N\} \quad (2)$$

The newly formed samples serve as the input data for the second layer. The meta-learner L_{new} is derived using a second-layer prediction algorithm. The training outcomes of the first-layer algorithm are fully used in the induction process of the second-level algorithm, enabling the latter to identify and minimize prediction errors from the first-layer algorithm. This iterative process contributes to improving the model's accuracy [30].

In this study, we begin with the first layer which is base learner, utilizing five distinct algorithms: Adaptive Boosting (AdaBoost), Gradient Boosting, K-Nearest Neighbor (KNN), Random Forest (RF), and Support Vector Machine (SVM). Each base learner algorithm undergoes cross-validation and parameter tuning to attain an optimal model. Then, the most optimal tuning outcomes from each algorithm are stacked, serving as the input for the second layer with Multi-Layer Perceptron (MLP) as a meta learner. The meta learner is then applied to the testing data.

D. Evaluation

Confusion matrix is used to measure performance of a model. This matrix provides insights into the predicted and actual classifications, helping in the assessment of classifier accuracy. The columns represent predicted events, while the rows correspond to actual events [31]. Model performance measurement in the confusion matrix include accuracy, precision, recall, and F1-score.

Precision is a measure of the actual positive value (true positive) against the total number of positive events, both actual and predicted results (true positive + false positive). Recall is a measure of the number of actual positive class predictions (true positive) against all actual events (true positive + false negative). F1-score is a measure that balances precision and recall. Accuracy is a measure of the comparison of all actual results to all results, both predicted and actual [32]. All of the measurement can be shown in equation (3) until (6):

$$\text{Accuracy} = \frac{\text{True Positive} + \text{True Negative}}{\text{True Positive} + \text{False Positive} + \text{True Negative} + \text{False Negative}} \quad (3)$$

$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}} \quad (4)$$

$$\text{Recall} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}} \quad (5)$$

$$\text{F1 Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (6)$$

IV. RESULT AND DISCUSSION

The raw datasets consist of 15235 rows approved drug data, 5346 rows approved protein data, 522 rows rejected drug data, and 241 rows rejected protein data (each drug and protein data have 23 and 13 columns respectively). These datasets have many missing values and identifier columns and need to be merged, so we went through data preprocessing step. Data preprocessing generates 5520 rows and 5 columns data. Then, data encoded to fit the stacking ensemble learning model, generates 5520 rows and 6861 columns data. Imbalanced experiment result data then oversampled with SMOTE. Synthetic data was added to the minor class data (specifically, 'Rejected'). The comparison of imbalanced and balanced data can be shown in Fig. 2.

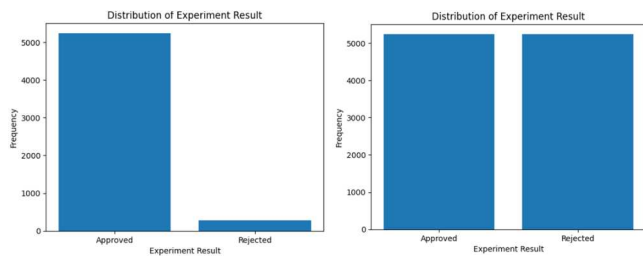


Fig. 2. Comparison of dataset before and after oversampling with SMOTE

The preprocessed dataset is inputted into five base learner models with basic tuning. Then, the models were evaluated using cross validation (CV) score, with the results presented in Table III.

TABLE III. BASE LEARNER MODEL'S CROSS VALIDATION SCORE

Base Learner Model	Cross Validation Score (Mean)
Adaptive Boosting	0.6709
Gradient Boosting	0.7128
K-Nearest Neighbor	0.5166
Random Forest	0.8294
Support Vector Machine	0.9130

Based on the CV score, SVM achieves the highest mean cross-validation score, indicating has the best generalization performance and effective separation of classes in the feature space. Random Forest follows closely behind, showing strong predictive capabilities and robustness across different subsets of the data. Gradient Boosting achieves a moderate mean cross-validation score, indicating good generalization performance and effective learning from the data. AdaBoost also performs moderately well, although it lags behind SVM and Random Forest. On the other hand, KNN achieves the lowest mean cross-validation score, suggesting weakest generalization performance and potential limitations among the other algorithms in capturing complex patterns in the data.

Because some of base models have weak performance, hyperparameter tuning was conducted to refine these models for optimal performance. Each parameter was tested with two different values for categorical value and three different values for numerical value to produce the best value.

TABLE IV. PARAMETERS USED IN HYPERPARAMETER TUNING

Model	Parameter	Value(s)	Best Value
AdaBoost	n-estimators	50, 100, 200	200
	learning rate	0.1, 0.01, 0.001	0.1
GBoost	n-estimators	50, 100, 200	200
	learning rate	0.1, 0.01, 0.001	0.1
KNN	n_neighbors	3, 5, 7	3
	weights	'uniform', 'distance'	'uniform'
RF	n-estimators	50, 100, 200	200
	max depth	3, 5, 7	7
SVM	C	0.1, 1, 10	10
	kernel	'linear', 'rbf'	'rbf'

As shown in Tabel IV, boosting models like AdaBoost and GBoost perform better with a higher number of estimators (200) and a learning rate of 0.1. Similarly, the Random Forest (RF) model benefits from deeper trees (max depth of 7) and more estimators (200). For Support Vector Machine (SVM), the best performance is achieved with a higher regularization parameter (C=10) and the 'rbf' kernel. Basic and fine tuning model then compared using CV score, with the results presented in Table V.

TABLE V. BASIC AND FINE TUNING MODEL COMPARISON

Model	Basic Tuning CV Score	Fine Tuning CV Score
AdaBoost	0.6709	0.8473
GBoost	0.7128	0.8516
KNN	0.5166	0.5277
RF	0.8294	0.8663
SVM	0.9130	0.9874

Based on Table V, fine tuning models generally resulted in significant improvements in CV score. The most notable improvements are seen in SVM and the boosting methods (AdaBoost and GBoost), which show substantial increases in their CV scores after fine tuning. However, KNN shows only a marginal improvement, suggesting that the tuning had less impact on its performance compared to the other models.

Each model with optimized parameters was then integrated into the stacking model with the MLP as the meta learner. Then, the stacking model was evaluated and has best performance with an accuracy of 99.047%. The comparison performance between single classifier and stacking ensemble classifier model followed by the confusion matrix of stacking model can be shown in Table VI and Fig. 3, respectively.

TABLE VI. EVALUATION OF METHODS USED FOR DTI ANALYSIS

Classifier	Accuracy (%)	Precision	Recall	F1-Score
AdaBoost	84.890%	0.990	0.700	0.820
GBoost	85.415%	0.984	0.715	0.828
KNN	53.289%	0.513	1.000	0.678
RF	85.272%	0.782	0.971	0.867
SVM	98.570%	0.997	0.973	0.985

Classifier	Accuracy (%)	Precision	Recall	F1-Score
Stacking	99.047%	0.993	0.988	0.991

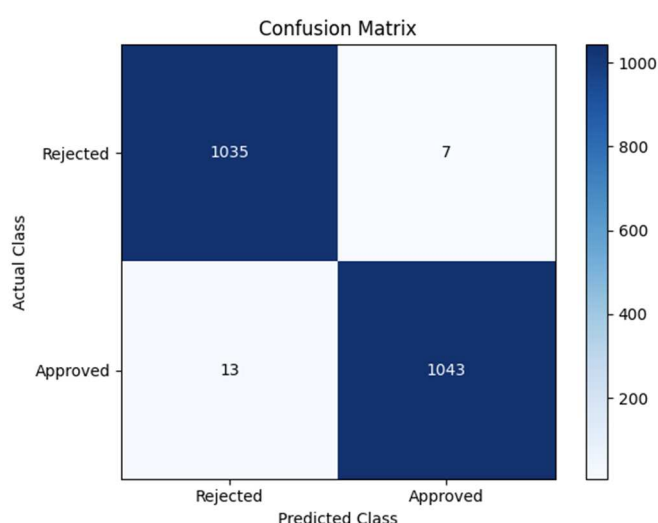


Fig. 3. Confusion Matrix of Stacking Model

Based on Fig. 3, we can conclude that the model shows a great ability to accurately classify a significant portion of DTI pairs, with a notable number of rejected (1035 true negatives) or approved (1043 true positives). This robust performance is indicative of the model's efficacy in discerning between positive and negative DTI outcomes.

However, despite its overall success, the stacking model does exhibit some misclassifications. Specifically, a small fraction of DTI pairs is wrongly classified as approved when they are actually rejected (7 false positives), and vice versa (13 false negatives). While these misclassifications are relatively minor in comparison to the total number of predictions, they highlight potential areas for improvement in the model's predictive capabilities.

V. CONCLUSION

This study introduces a novel approach to drug-target interaction (DTI) prediction by utilizing stacking ensemble techniques, with a Multi-Layer Perceptron (MLP) as the meta learner. By combining different base learners and optimizing their performance through the meta learner, the model demonstrates remarkable efficacy in accurately predicting DTI classifications. Additionally, the integration of the SMOTE oversampling method effectively addresses imbalanced data challenges, ensuring robust predictions across all classes. The evaluation of the model resulted in an exceptional accuracy rate of 99.047%, highlighting the potential of ensemble learning techniques in advancing DTI prediction capabilities.

Future research initiatives could delve into several key areas to further enhance the model's performance and applicability. One promising direction is the exploration of alternative oversampling techniques beyond SMOTE. While SMOTE has proven effective in addressing data imbalance, other methods such as ADASYN (Adaptive Synthetic Sampling) or borderline-SMOTE could be investigated to generate more representative synthetic samples. These techniques may improve the model's ability to handle more complex DTI datasets, leading to even better predictive performance.

Another important avenue for future research involves experimenting with different meta learners. Although the MLP showed promising results as a meta learner in this study, future investigations could consider alternative meta learners like XGBoost, LightGBM, or deep learning architectures such as convolutional neural networks (CNNs). Comparing these options could yield valuable insights into further optimizing the model's prediction accuracy and robustness. Additionally, incorporating advanced feature selection methods, such as Principal Component Analysis (PCA), may help reduce the dimensionality of the dataset, potentially improving both the efficiency and accuracy of the model.

Beyond these technical enhancements, future studies could explore different ensemble strategies, including bagging, boosting, or hybrid approaches, to further strengthen the model's generalization capabilities. Applying the model to diverse datasets, including other types of biological interactions or different drug-target databases, could help validate its generalizability and robustness. Finally, real-world implementation and validation in clinical settings should be prioritized. Collaborations with pharmaceutical companies or research institutions could facilitate the application of the model to ongoing drug discovery projects, providing practical feedback and opportunities for further refinement.

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